QUERY PROCESSING LAB EXP 20 to 30

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COURSE CODE : DSA0503 - Query Processing

EXP 21

Aim

To write a Pandas program to swap the cases of a specified character column in a given DataFrame.

Algorithm

- Import Pandas.
- Create a DataFrame containing a character/string column.
- Select the required column.
- Apply the swapcase() function.
- Display the original and modified DataFrame.

Code

```
import pandas as pd

data = {
    'Name': ['Udhaya', 'Kumar', 'Python', 'DataScience']
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

df['Name'] = df['Name'].str.swapcase()

print("\nAfter Swapping Cases:")
print(df)
```

Output

```
>>>
===== RESTART: C:/Users/udhay/AppData/Local/Programs/Python/Python311/m.py =====
Original DataFrame:
          Name
0       Udhaya
1        Kumar
2       Python
3  DataScience

After Swapping Cases:
          Name
0       uDHAYA
1        kUMAR
2       pYTHON
3  dATAsCIENCE
>>>
```

EXP 22

Aim

To write a Python program to draw a line with suitable labels on the X-axis, Y-axis and title.

Algorithm

- Import matplotlib.pyplot.
- Define X-axis values.
- Define Y-axis values.
- Plot the values using plt.plot().
- Add X-axis label.
- Add Y-axis label.
- Add title.
- Display the graph.

Code

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

plt.plot(x, y)

plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Simple Line Graph")

plt.show()
```

Output

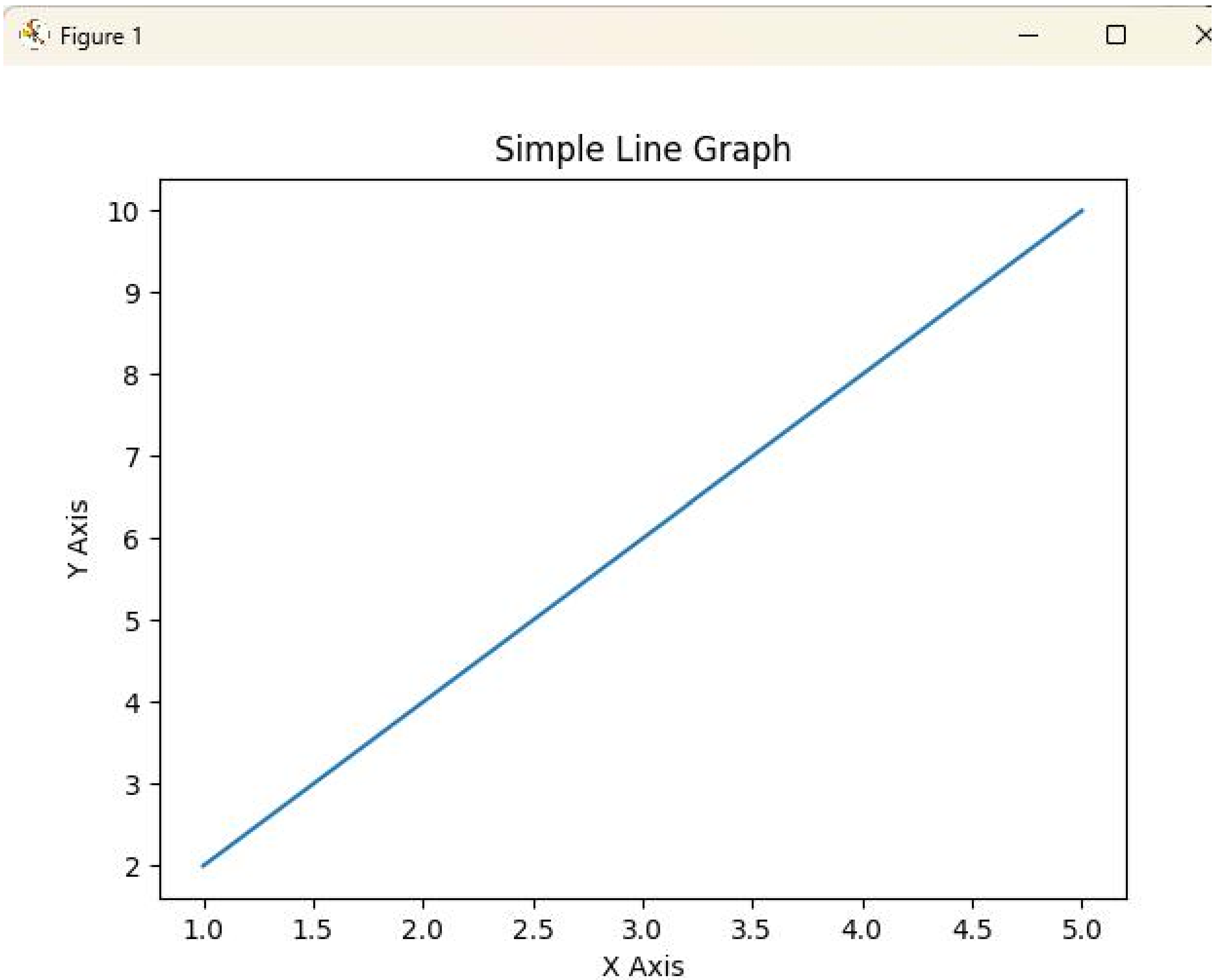


Figure 1
Simple Line Graph
10
9
8
7
6
5
4
3
2
Y Axis
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
X Axis

EXP 23

Aim

To write a Python program to draw a line using axis values taken from a text file, with suitable X-axis label, Y-axis label and title. The supplied test file contains 1 2, 2 4, and 3 1.

Algorithm

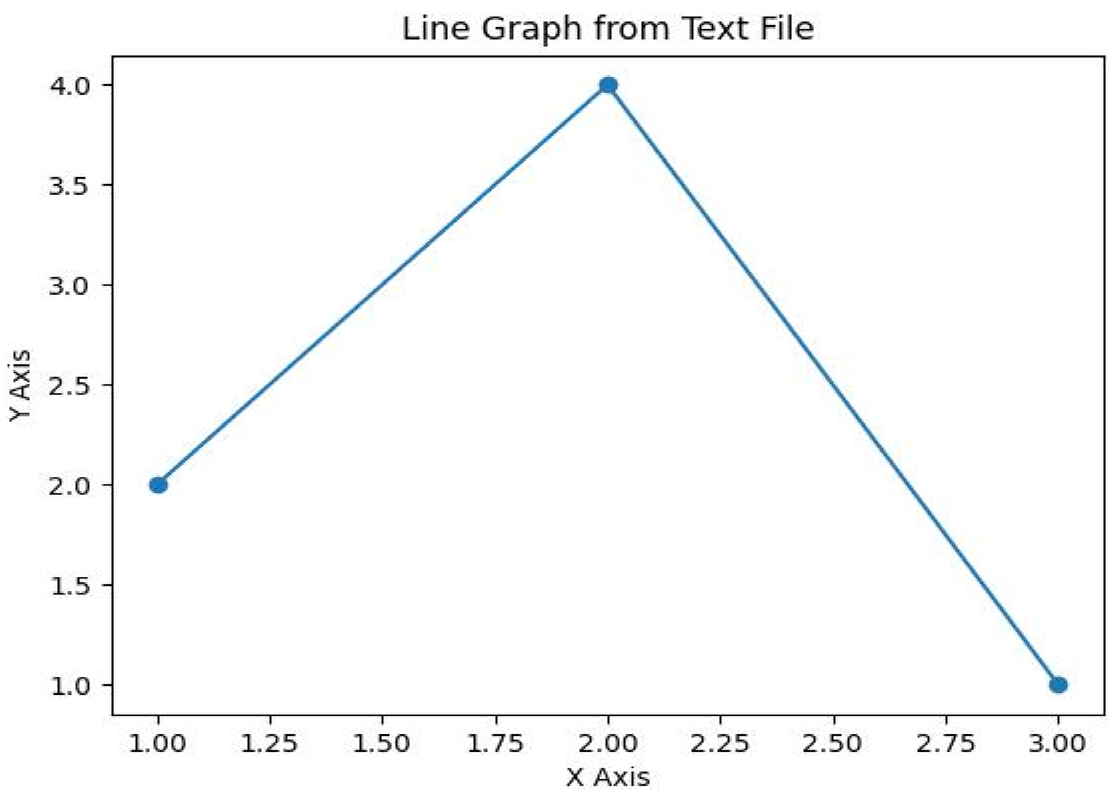
- Create/read the test.txt file.
- Read X and Y values from the file.
- Store X values and Y values separately.
- Plot the values.
- Add labels and title.
- Display the graph.

Code

```
import matplotlib.pyplot as plt

# Create test.txt
with open("test.txt", "w") as file:
    file.write("1 2\n")
    file.write("2 4\n")
    file.write("3 1\n")

x = []
y = []

with open("test.txt", "r") as file:
    for line in file:
        values = line.split()
```

```
        x.append(float(values[0]))
        y.append(float(values[1]))

plt.plot(x, y, marker='o')

plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Line Graph from Text File")

plt.show()
```

Output



EXP 24

Aim

To write a Python program to draw line charts of the financial data of Alphabet Inc. between October 3, 2016 and October 7, 2016.

Algorithm

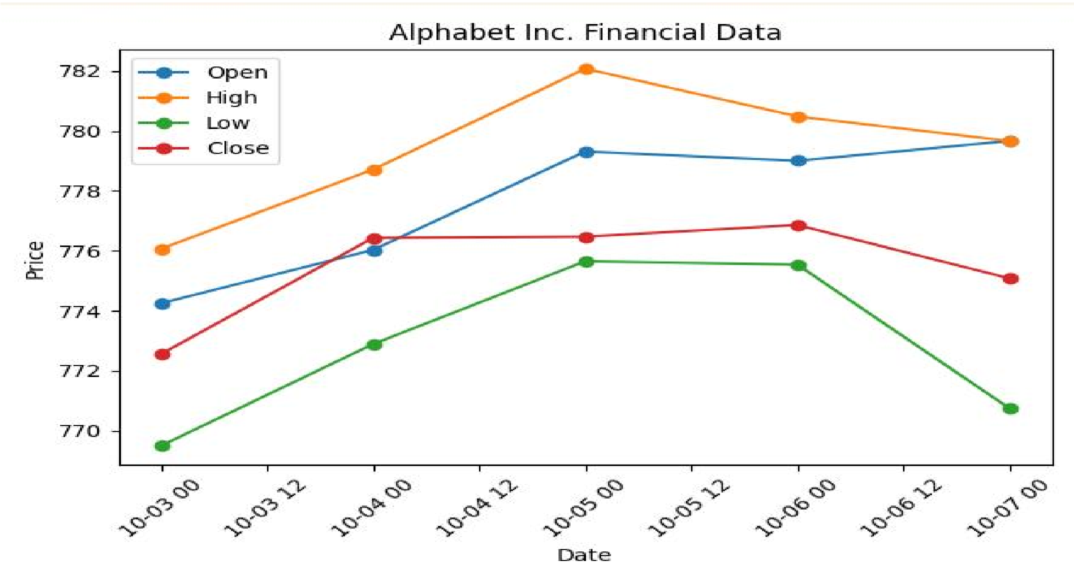
- Import Pandas and Matplotlib.
- Create the financial data.
- Convert the Date column into date format.
- Plot Open, High, Low and Close prices.
- Add title and axis labels.
- Display the graph.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Date': ['10-03-16', '10-04-16', '10-05-16',
             '10-06-16', '10-07-16'],
    'Open': [774.25, 776.030029, 779.309998, 779, 779.659973],
    'High': [776.065002, 778.710022, 782.070007, 780.47998,
779.659973],
    'Low': [769.5, 772.890015, 775.650024, 775.539978, 770.75],
    'Close': [772.559998, 776.429993, 776.469971,
              776.859985, 775.080017]
}

df = pd.DataFrame(data)
```

```
df['Date'] = pd.to_datetime(df['Date'])

plt.plot(df['Date'], df['Open'], marker='o', label='Open')
plt.plot(df['Date'], df['High'], marker='o', label='High')
plt.plot(df['Date'], df['Low'], marker='o', label='Low')
plt.plot(df['Date'], df['Close'], marker='o', label='Close')

plt.xlabel("Date")
plt.ylabel("Price")
plt.title("Alphabet Inc. Financial Data")

plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Output



EXP 25

Aim

To write a Python program to plot two or more lines with legends, different widths and colors.

Algorithm

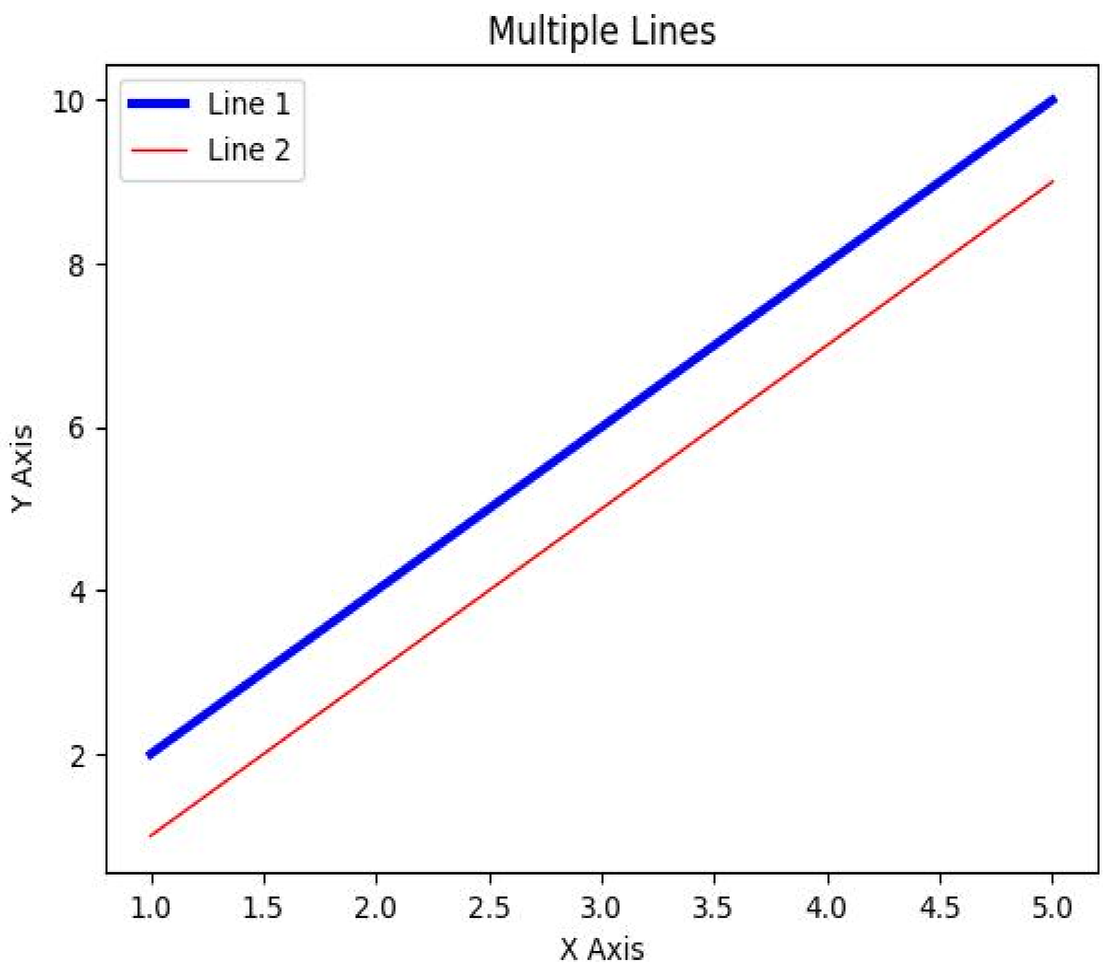
- ➢ Import Matplotlib.
- ➢ Define X-axis values.
- ➢ Define two sets of Y values.
- ➢ Plot the first line.
- ➢ Plot the second line.
- ➢ Specify line width and color.
- ➢ Add legends.
- ➢ Display the graph.

Code

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]

y1 = [2, 4, 6, 8, 10]
y2 = [1, 3, 5, 7, 9]

plt.plot(x, y1, color='blue', linewidth=3, label='Line 1')
plt.plot(x, y2, color='red', linewidth=1, label='Line 2')

plt.xlabel("X Axis")
plt.ylabel("Y Axis")
```

```
plt.title("Multiple Lines")

plt.legend()
plt.show()
```

Output



EXP 26

Aim

To write a Python program to create multiple plots.

Algorithm

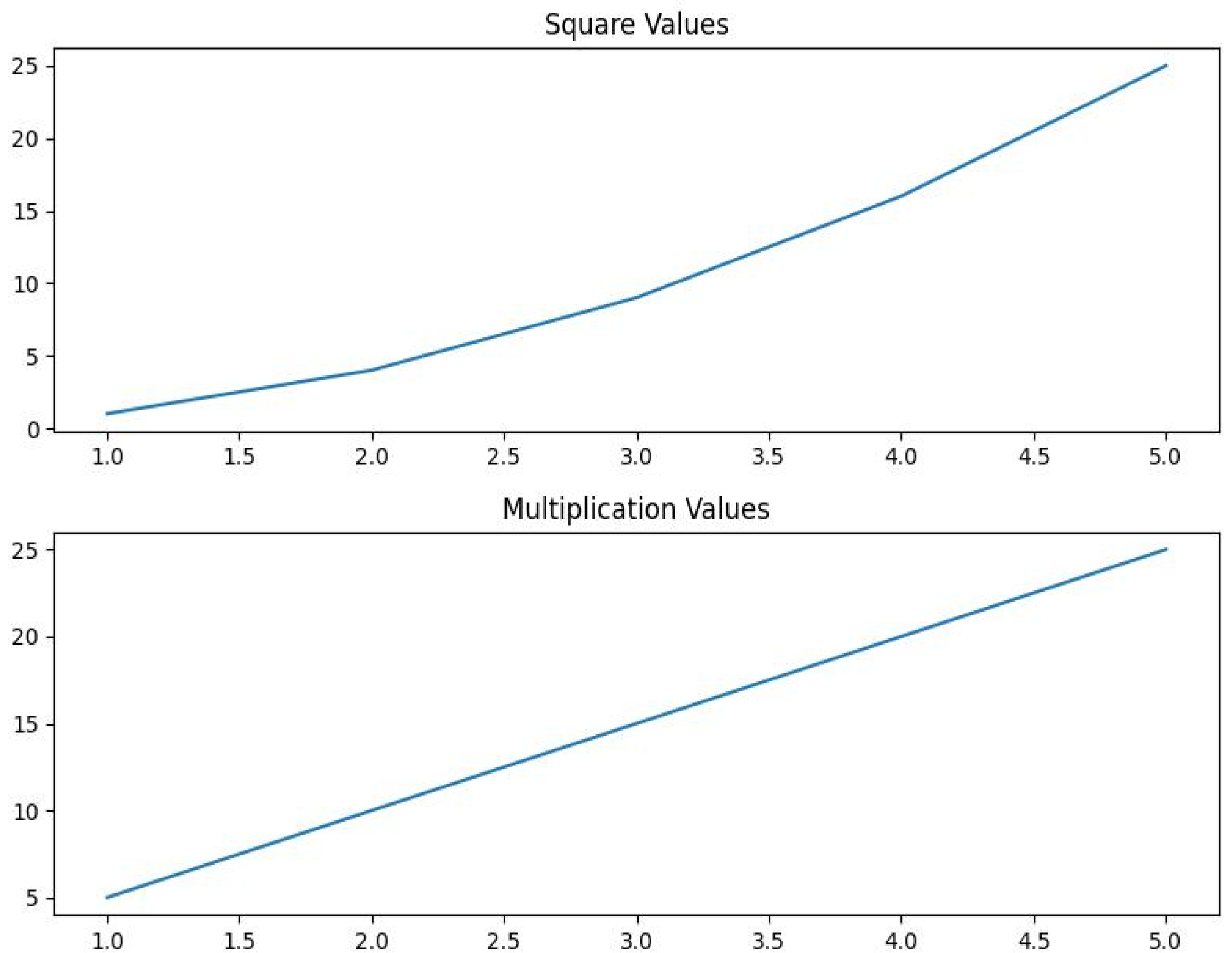
1. Import Matplotlib.
2. Create X and Y values.
3. Create multiple subplots.
4. Plot different graphs in each subplot.
5. Give suitable titles.
6. Display all plots.

Code

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]

y1 = [1, 4, 9, 16, 25]
y2 = [5, 10, 15, 20, 25]

plt.figure(figsize=(8, 6))

plt.subplot(2, 1, 1)
plt.plot(x, y1)
plt.title("Square Values")

plt.subplot(2, 1, 2)
plt.plot(x, y2)
plt.title("Multiplication Values")
```

```
plt.tight_layout()
plt.show()
```

Output



EXP 27

Aim

To write a Python program to display a bar chart showing the popularity of programming languages. The experiment specifies Java, Python, PHP, JavaScript, C# and C++ with popularity values 22.2, 17.6, 8.8, 8, 7.7 and 6.7.

Algorithm

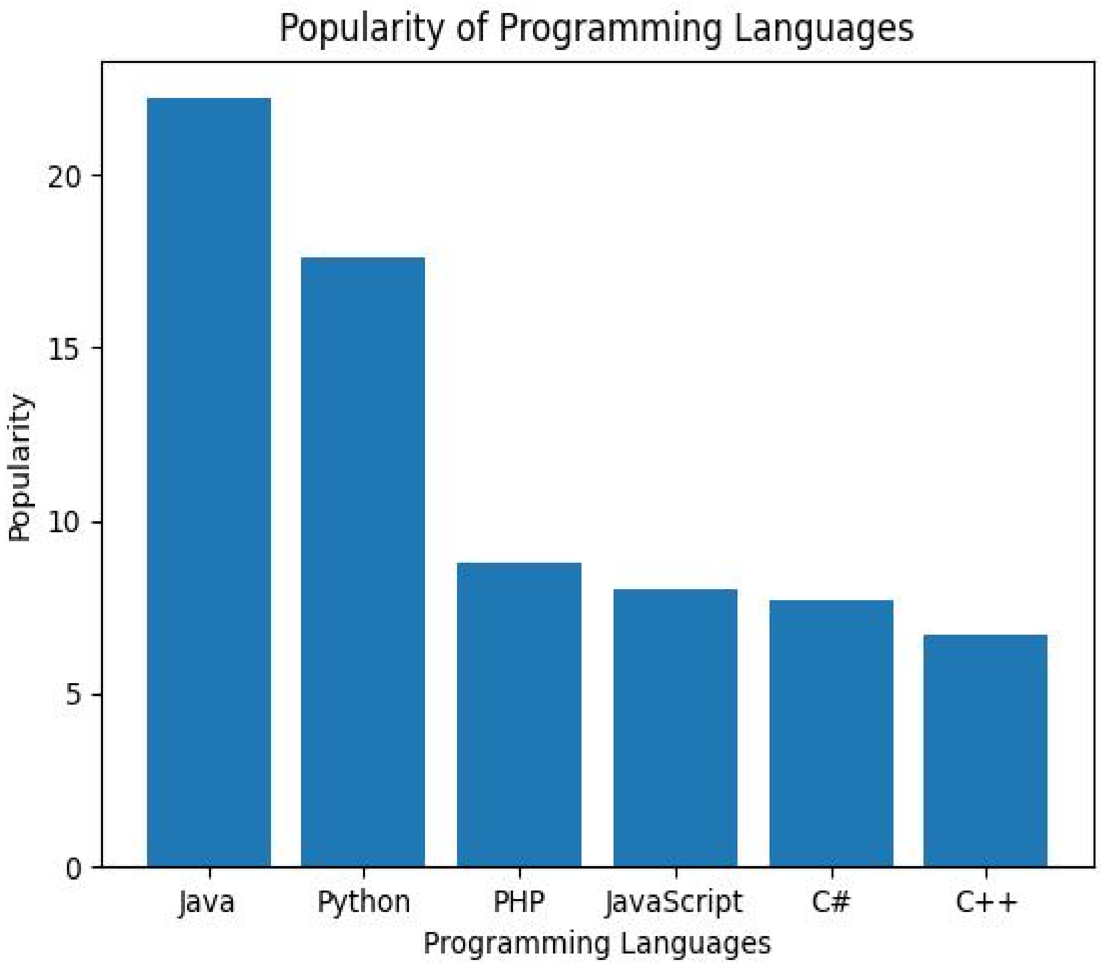
- Import Matplotlib.
- Create a list of programming languages.
- Create their popularity values.
- Create a bar chart.
- Add X-axis and Y-axis labels.
- Add title.
- Display the chart.

Code

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.bar(languages, popularity)

plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")
```

```
plt.show()
```

Output



EXP 28

Aim

To write a Python program to display a horizontal bar chart of the popularity of programming languages.

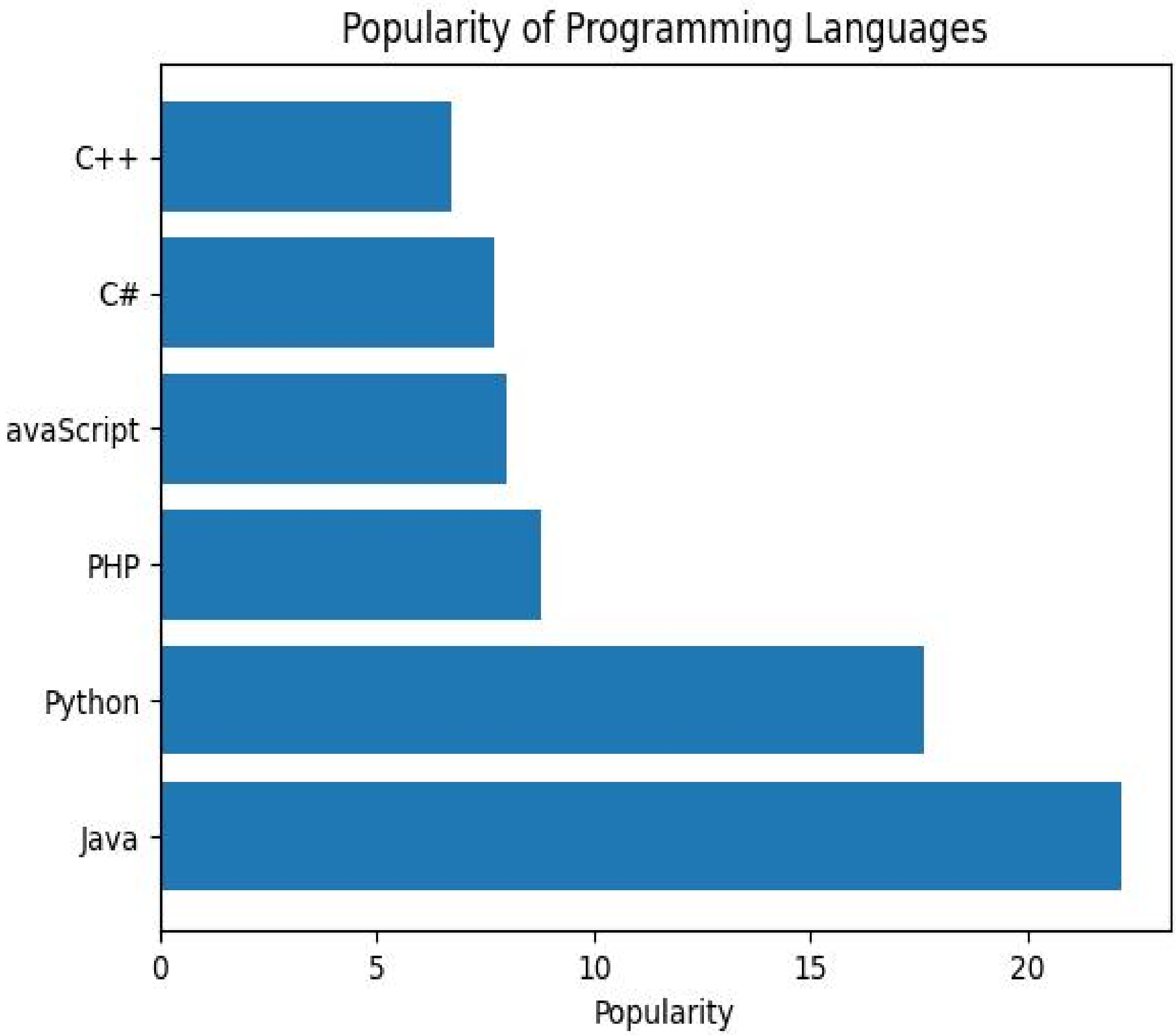
Algorithm

- Import Matplotlib.
- Define programming languages.
- Define popularity values.
- Use barh() to create a horizontal bar chart.
- Add labels and title.
- Display the chart.

Code

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

plt.barh(languages, popularity)

plt.xlabel("Popularity")
plt.ylabel("Programming Languages")
plt.title("Popularity of Programming Languages")

plt.show()
```

Output


Popularity of Programming Languages
C++
C#
avaScript
PHP
Python
Java
0
5
10
15
20
Popularity

EXP 29

Aim

To write a Python program to display a bar chart of programming language popularity using different colors for each bar.

Algorithm

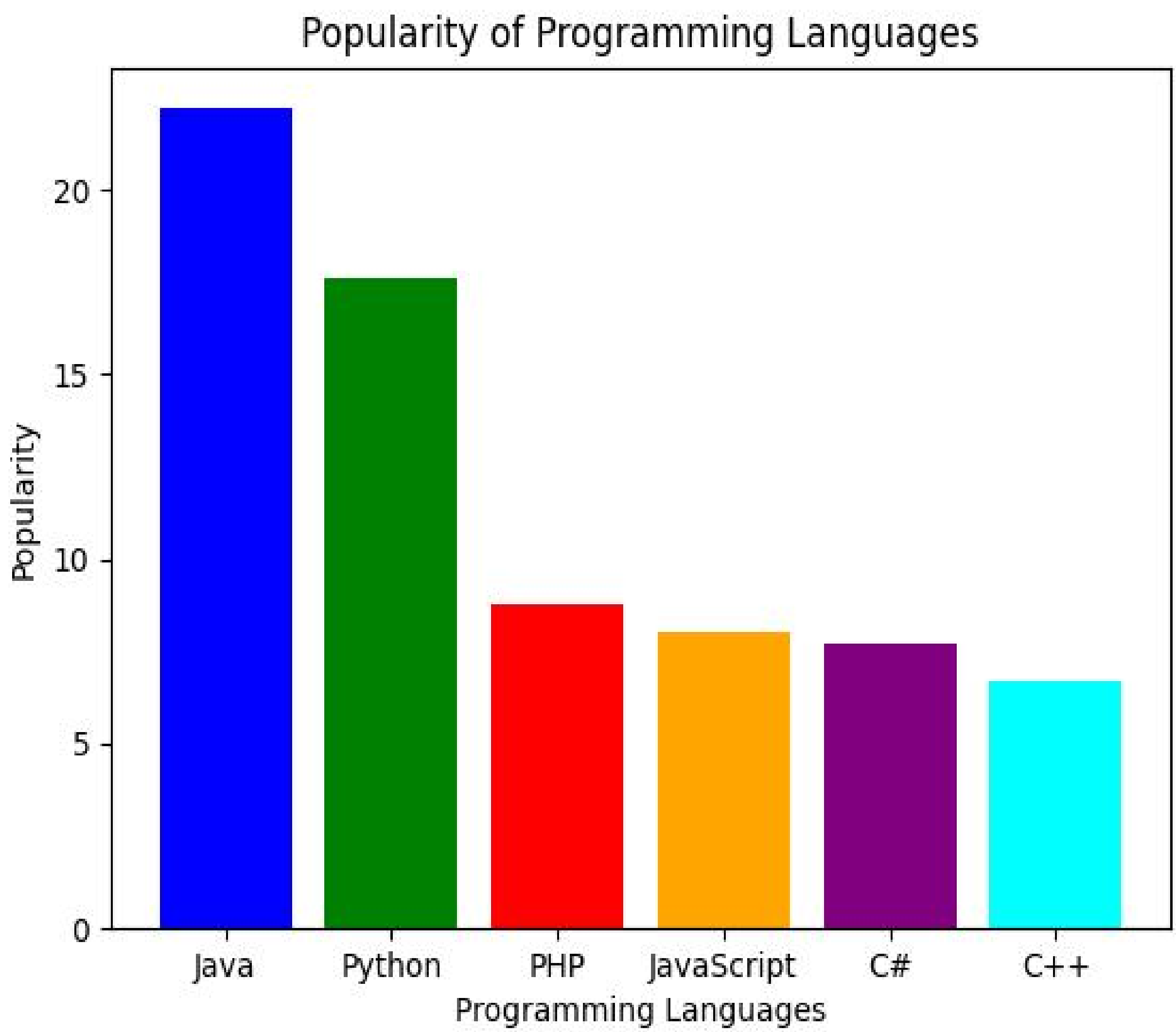
- Import Matplotlib.
- Define programming languages.
- Define popularity values.
- Define different colors.
- Create the bar chart.
- Add labels and title.
- Display the graph.

Code

```
import matplotlib.pyplot as plt

languages = ['Java', 'Python', 'PHP', 'JavaScript', 'C#', 'C++']
popularity = [22.2, 17.6, 8.8, 8, 7.7, 6.7]

colors = ['blue', 'green', 'red', 'orange', 'purple', 'cyan']

plt.bar(languages, popularity, color=colors)

plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")
```

```
plt.show()
```

Output



EXP 30

Aim

To write a Python program to create a bar plot of scores by group and gender using multiple X values for men and women. The given sample data are Men = (22, 30, 35, 35, 26) and Women = (25, 32, 30, 35, 29).

Algorithm

- Import Matplotlib and NumPy.
- Define group numbers.
- Define men's scores.
- Define women's scores.
- Create X positions.
- Shift the positions of men and women bars.
- Draw the bars.
- Add labels, title and legend.
- Display the chart.

Code

```
import numpy as np
import matplotlib.pyplot as plt

groups = ['Group 1', 'Group 2', 'Group 3', 'Group 4', 'Group 5']

men = [22, 30, 35, 35, 26]
women = [25, 32, 30, 35, 29]

x = np.arange(len(groups))
```

```
width = 0.35

plt.bar(x - width/2, men, width, label='Men')
plt.bar(x + width/2, women, width, label='Women')

plt.xlabel("Groups")
plt.ylabel("Scores")
plt.title("Scores by Group and Gender")

plt.xticks(x, groups)
plt.legend()

plt.show()
```

OUTPUT